

Lesson 60 Practice: Current OS, Processes, Linux and Utilities

JKSSB Computer Awareness

October 8, 2026

Provenance: 18 original JKSSB-pattern questions (*[Original]*) and two paraphrased adaptations of the JKSSB Junior Assistant 2026 computer block (Q70 deadlock recovery by resource preemption, and Q71 priority inversion/inheritance; final key). The adaptations at Q1 and Q3 are reworded, not verbatim reproductions; no item in this set reproduces an official paper item verbatim.

- Q1.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q70; final key; paraphrased]* A system is in a deadlock. Which method recovers from the deadlock by taking a resource away from one process and giving it to another?
- A. Resource preemption
 - B. Mutual exclusion
 - C. Hold and wait
 - D. Circular wait
- Q2.** *[Original]* Which of the following is **NOT** one of the four necessary conditions for deadlock?
- A. Mutual exclusion
 - B. Hold and wait
 - C. Preemption
 - D. Circular wait
- Q3.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q71; final key; paraphrased]* A low-priority task holds a resource that a high-priority task needs, and meanwhile a medium-priority task preempts the low-priority task and runs. This situation is called:
- A. priority inversion
 - B. priority inheritance
 - C. deadlock
 - D. fragmentation
- Q4.** *[Original]* Which technique resolves priority inversion by temporarily raising the priority of the task that holds the needed resource?
- A. Time slicing
 - B. Priority inheritance
 - C. Spooling
 - D. Paging

- Q5.** *[Original]* Which of the following does **NOT** by itself prevent priority inversion?
- A. Priority inheritance
 - B. Priority ceiling protocol
 - C. Round-robin scheduling
 - D. Priority boosting
- Q6.** *[Original]* A ready process is repeatedly passed over and waits for an indefinite time because higher-priority processes keep arriving. This condition is called:
- A. starvation
 - B. deadlock
 - C. thrashing
 - D. fragmentation
- Q7.** *[Original]* Which scheduling technique gradually increases the priority of a long-waiting process so that it eventually runs?
- A. Aging
 - B. Swapping
 - C. Spooling
 - D. Caching
- Q8.** *[Original]* In preemptive priority scheduling, the CPU is taken away from the running process when:
- A. the running process finishes only.
 - B. a process with a higher priority becomes ready.
 - C. the running process performs an input/output operation only.
 - D. the running process requests more memory only.
- Q9.** *[Original]* Expand RTOS.
- A. Real-Time Operating System
 - B. Rapid Task Operating System
 - C. Remote Terminal Operating System
 - D. Read-Only Operating System
- Q10.** *[Original]* Evaluate the statements:
- I. In priority inheritance, a low-priority task holding a needed resource temporarily inherits the priority of the high-priority task waiting for it.
 - II. Round-robin scheduling by itself prevents priority inversion.
 - III. Starvation means a process may wait indefinitely while other processes are repeatedly served.
- A. I and II only

- B. II and III only
 - C. I and III only
 - D. I, II, and III
- Q11.** *[Original]* The core part of the Linux operating system, which manages processes, memory, and devices, is called the:
- A. Linux kernel
 - B. Linux shell
 - C. GNOME desktop
 - D. GRUB bootloader
- Q12.** *[Original]* In Linux, which pairing of a directory with its purpose is correct?
- A. `/root` — the home directory of the superuser (root user)
 - B. `/home` — the home directory of the superuser
 - C. `/root` — the home directories of ordinary users
 - D. `/home` — the root of the whole file system
- Q13.** *[Original]* Which file system is native to Linux and is commonly its default?
- A. NTFS
 - B. ext4
 - C. FAT32
 - D. exFAT
- Q14.** *[Original]* Which file system is the default for modern Windows installations?
- A. ext4
 - B. NTFS
 - C. ext3
 - D. HFS+
- Q15.** *[Original]* What is the Windows Store (Microsoft Store)?
- A. A file compression utility
 - B. A disk defragmentation tool
 - C. An online marketplace used to download and install apps for Windows
 - D. A Linux package manager
- Q16.** *[Original]* Which statement correctly contrasts Notepad and WordPad?
- A. Notepad supports rich text formatting, while WordPad is plain text only.
 - B. Notepad is a plain-text editor, while WordPad supports basic formatting such as bold and font styles.
 - C. Both are spreadsheet programs.

- D. WordPad cannot save a file in any format.
- Q17.** *[Original]* Which built-in Windows utility deletes unnecessary files such as temporary files and the Recycle Bin contents to free disk space?
- A. Disk Cleanup
 - B. Disk Defragmenter
 - C. Device Manager
 - D. Task Scheduler
- Q18.** *[Original]* Why should a solid-state drive (SSD) generally **NOT** be defragmented?
- A. It has no moving parts, so defragmenting gives no speed benefit and only adds unnecessary write wear; TRIM is used instead.
 - B. It is faster than RAM and does not need any maintenance.
 - C. It is volatile and loses data when defragmented.
 - D. It stores data only in RAM.
- Q19.** *[Original]* Which Windows utility is used to scan documents and send or receive faxes (the imaging and fax tool)?
- A. Windows Fax and Scan
 - B. WordPad
 - C. Disk Cleanup
 - D. Windows Store
- Q20.** *[Original]* What does the command `xcopy /s` do?
- A. Copies files and subdirectories, but excludes empty ones.
 - B. Deletes files and subdirectories permanently.
 - C. Renames a single file only.
 - D. Formats a disk.